RPractical

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PART A

```
# A1
v <- seq(5, 50, by = 5)
print(length(v))
```

```
## [1] 10
```

```
print(sum(v))
```

```
## [1] 275
```

```
print(mean(v))
```

```
## [1] 27.5
```

```
# A2
w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
print(w)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

```
# A3
print(17 %% 5)       # 2
```

```
## [1] 2
```

```
print(17 %/% 5)      # 3
```

```
## [1] 3
```

```r
print(2^10)
```

```
## [1] 1024
```

```r
print(sqrt(144))
```

```
## [1] 12
```

```r
print(round(pi, 3))
```

```
## [1] 3.142
```

```r
# A4
a <- c(2, 4, 6, 8); b <- c(1, 3, 5, 7)
print(a + b)
```

```
## [1]  3  7 11 15
```

```r
print(a * b)
```

```
## [1]  2 12 30 56
```

```r
print(a > b)
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
print(sum(a * b))
```

```
## [1] 100
```

```r
# A5
print(paste("Student", 1:3, sep = '-'))
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
print(class(42))
```

```
## [1] "numeric"
```

```r
print(class(42L))
```

```
## [1] "integer"
```

```r
print(class("42"))
```

```
## [1] "character"
```

```r
print(class(41i))
```

```
## [1] "complex"
```

```r
print(class(TRUE))
```

```
## [1] "logical"
```

PART B

```r
# B1
students <- data.frame(
  id = 1:8,
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
```

```r
# B2
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```r
print(dim(students))
```

```
## [1] 8 7
```

```r
print(names(students))
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```r
print(summary(students))
```

```
##        id              name               sex                section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

```r
# B3
students$average <- round((students$midterm + students$final) / 2, 1)
```

```r
# B4
students$passed <- students$average >= 70 & students$attendance >= 75
```

```r
# B5
students$standing <- ifelse(students$average >= 90, "Honors",
                     ifelse(students$average >= 75, "Good", "Needs Support"))
```

```r
# B6
classify <- function(score) {
  if (score >= 85) {
    "Distinction"
  } else if (score >= 70) {
    "Merit"
  } else if (score >= 50) {
    "Pass"
  } else {
    "Fail"
  }
}
print(sapply(c(90, 73, 55, 40), classify))  # "Distinction" "Merit" "Pass" "Fail"
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

PART C

```r
# C1
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
```

```
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 70),
  stringsAsFactors = FALSE
)
new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                         ifelse(new_students$average >= 75, "Good", "Needs Support"))

students_all <- rbind(students, new_students)
print(students_all)
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         70    69.0  FALSE Needs Support
```

```
# C2
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
print(class(students_all))   # "data.frame"
```

```
## [1] "data.frame"
```

```
# C3
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
print(m)
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

```r
# C4
# rbind() fails with "names do not match previous names" when the data
# frames have different column names, because it can't line up the columns.
# tryCatch() lets us show the message instead of crashing.
bad <- students
names(bad)[2] <- "nm"
tryCatch(rbind(students, bad),
         error = function(e) { message("rbind failed: ", conditionMessage(e)); NULL })
```

```
## rbind failed: names do not match previous names
```

```
## NULL
```

PART D

```r
# D1
print(students_all$name)
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
print(students_all[["name"]])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
print(students_all[, "name"])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
# D2
print(students_all[2:5, c("name", "midterm", "final")])
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

```r
# D3
print(students_all[students_all$attendance > 90, ])
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

```r
# D4
print(subset(students_all,
             sex == "F" & section %in% c("A", "C") & final >= 75,
             select = c(name, section, final)))
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

```r
# D5
print(students_all$name[which.max(students_all$average)])
```

```
## [1] "Chen"
```

```r
print(students_all$name[which.min(students_all$average)])
```

```
## [1] "Hiro"
```

```r
# D6
sorted <- students_all[order(students_all$average, decreasing = TRUE), ]
print(head(sorted, 3))
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

PART E

```r
# E1
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)
print(levels(students_all$section))
```

```
## [1] "A" "B" "C"
```

```r
print(levels(students_all$sex))
```

```
## [1] "F" "M"
```

```r
# E2
print(table(students_all$sex))
```

```
##
## F M
## 5 5
```

```r
print(table(students_all$section))
```

```
##
## A B C
## 4 3 3
```

```r
print(table(students_all$sex, students_all$section))
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

```r
#    A B C
# F  2 1 2
# M  2 2 1
```

```r
# E3
print(as.integer(students_all$sex))
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```r
# The integers are the level codes: 1 = "F", 2 = "M" (levels are sorted alphabetically).
```

```r
# E4
sex_label <- factor(students_all$sex, levels = c("F", "M"),
                    labels = c("Female", "Male"))
students_all$sex_label <- sex_label
print(levels(students_all$sex_label))
```

```
## [1] "Female" "Male"
```

```r
# E5
print(summary(students_all$sex_label))
```

```
## Female   Male
##      5      5
```

```r
print(mean(students_all$average[students_all$sex_label == "Female"]))
```

```
## [1] 80.5
```

```r
print(mean(students_all$average[students_all$sex_label == "Male"]))
```

```
## [1] 79.3
```